

Riemann Stereographic Energy Coupling Suggests Motor-Imagery-Specific EEG Network Reconfiguration

Bayazit Karaman

Department of Computer Science

Florida Polytechnic University

Lakeland, FL, USA

bkaraman@floridapoly.edu

Abstract—Electroencephalography (EEG) provides millisecond-scale access to neural dynamics, but extracting task-sensitive network features from scalp recordings remains challenging. Sensor-level coupling can be affected by nonstationarity, volume conduction, reference choice, and ongoing rhythmic activity. This paper introduces Riemann Stereographic Energy Coupling (RSEC), a nonlinear feature-extraction method for detecting transient shared-energy events across EEG channel pairs. RSEC transforms each EEG channel into an analytic signal, normalizes it within each epoch, maps it to a Riemann stereographic energy representation, and detects shared-energy events through fixed-precision channel-pair matching. Pairwise event matrices, node participation scores, and global RSEC density were evaluated using the public PhysioNet EEG Motor Movement/Imagery dataset. Of the 109 available subjects, three were excluded because their baseline and task recordings had inconsistent sampling frequencies, leaving 106 subjects for paired analysis. Motor imagery significantly reduced global RSEC density relative to rest, whereas motor execution did not produce a significant global-density change. The imagery-related task-minus-rest effect also differed significantly from both the execution effect and phase-randomized surrogate EEG. These findings suggest that RSEC captures motor-imagery-related changes in sensor-space EEG energy organization.

Index Terms—EEG, motor imagery, motor execution, Riemann stereographic projection, Riemann Stereographic Energy Coupling, shared-energy events, biomedical signal processing, brain connectivity

I. INTRODUCTION

Electroencephalography (EEG) is widely used in biomedical signal processing because it records brain activity noninvasively with high temporal resolution. This temporal resolution is especially important in motor imagery and brain-computer interface (BCI) studies, where task-related modulation of sensorimotor rhythms can occur within short time intervals [1], [2]. Despite this advantage, network-level EEG interpretation remains difficult. Scalp channels record mixtures of neural and non-neural sources, and apparent sensor-level coupling can be shaped by common-reference effects, volume conduction, and ongoing rhythmic activity [3]–[5]. Task-sensitive signal summaries are therefore needed, but they should not equate sensor-level coupling with direct anatomical connectivity.

Motor imagery provides a useful setting for evaluating such features. Imagined movement recruits sensorimotor processes without overt movement and remains central to BCI and neurorehabilitation applications [2], [6]. Motor imagery and motor execution are related but not identical. Execution includes overt motor output and proprioceptive or sensory feedback, whereas imagery depends more strongly on internally generated motor planning with suppressed action. A feature that responds differently to imagery and execution may therefore capture motor-state-specific reconfiguration rather than a generic rest-to-task change.

Network neuroscience commonly uses graph-theoretic approaches to study functional integration, segregation, modularity, and hub organization [7]–[9]. EEG network studies often begin with pairwise measures such as coherence, phase-locking value (PLV), phase-lag index (PLI), weighted PLI, or imaginary coherence [4], [5], [10], [11]. These measures are valuable, but each emphasizes a particular form of association. Here, we evaluate a geometry-inspired alternative based on analytic-signal energy, Riemann stereographic projection, and fixed-precision shared-energy matching.

The proposed method, Riemann Stereographic Energy Coupling (RSEC), builds on earlier work on antipodal connectivity structures and Antipodal Hub Nodes (AHNs), where EEG signals were transformed into a complex-domain representation and analyzed through projection- and Riemann-based normalized transformations [12], [13]. The present formulation is intentionally conservative. Rather than interpreting detected events as anatomical hubs, RSEC treats repeated similarity in Riemann stereographic energy as operational sensor-space shared-energy events. Global RSEC density then summarizes how often these events occur across channel pairs and time samples.

We evaluate RSEC on the public PhysioNet EEG Motor Movement/Imagery dataset [14]–[16]. The main contributions are as follows: (1) formulation of RSEC as a reproducible EEG feature-extraction method; (2) conservative reframing of AHN-related ideas as operational sensor-space shared-energy features; (3) evaluation on 106 usable subjects from a public 64-channel EEG dataset; (4) demonstration of a motor-

imagery-specific reduction in global RSEC density; and (5) use of phase-randomized surrogate EEG to test whether the imagery effect is reproduced after phase disruption.

II. RELATED WORK

Motor imagery refers to the internal simulation of movement without overt motor output. EEG studies have consistently shown that imagined movement modulates ongoing sensorimotor rhythms, particularly through event-related desynchronization and synchronization in the alpha/mu and beta frequency bands [1], [2]. Although motor imagery and motor execution engage overlapping sensorimotor processes, execution also involves peripheral muscle activation, proprioceptive feedback, and sensory consequences associated with actual movement. Neuper et al. further showed that imagery mode can influence the spatial distribution of EEG activity, with kinesthetic motor imagery producing activity patterns localized near sensorimotor hand areas [6]. These findings motivate direct comparison of motor imagery and motor execution within the same EEG dataset.

Functional connectivity analysis provides a framework for quantifying statistical dependencies among signals recorded from different EEG sensors or brain regions. Common EEG pairwise measures include coherence, PLV, imaginary coherence, PLI, and wPLI [4], [5], [10], [11]. PLV measures the consistency of phase differences between signals over time or across trials and has been widely used to study EEG synchronization. However, phase-based connectivity measures, especially those sensitive to zero-lag synchrony, can be influenced by common sources, reference effects, and volume conduction. Imaginary coherence, PLI, and wPLI address part of this concern by emphasizing non-zero-lag interactions and reducing sensitivity to instantaneous coupling [4], [5], [11].

Although these measures are well established, they characterize specific forms of phase-based association and may not capture all task-sensitive structure in EEG signals. Nonlinear or geometry-inspired features may therefore provide complementary information about sensor-space EEG dynamics, provided their interpretation remains cautious. In this context, RSEC is evaluated as an operational sensor-space feature rather than as direct evidence of anatomical connectivity.

Brain network hubs are typically defined using graph measures such as degree, strength, centrality, or participation across modules [8], [9]. In scalp EEG, the term “hub” requires caution because sensors are not anatomical regions and edges depend on the selected coupling metric. For this reason, this paper uses the term RSEC shared-energy event for detected pairwise energy coincidences. AHN terminology is retained only as a historical link to prior work [12], [13]. High participation of a channel in RSEC events indicates repeated sensor-level involvement in transient energy coupling, not direct anatomical hub status.

Stereographic projection maps points from the extended complex plane onto the Riemann sphere and provides a geometric relationship between complex magnitude and chordal distance [17], [18]. RSEC applies this idea after converting

EEG time series into analytic signals. For each EEG channel, the Hilbert transform produces a complex analytic representation. After epoch-wise normalization, a chordal-distance energy value is computed for each channel and time sample. Channel pairs with matching energy at a given time point are counted as transient RSEC shared-energy events. The original AHN framework emphasized antipodal connectivity and increased event counts during active engagement [12]. The present study tests a broader and more falsifiable interpretation: RSEC density reflects state-dependent reconfiguration in sensor-space EEG energy coupling, and the direction of this change may vary across task states.

III. MATERIALS AND METHODS

A. Dataset and Preprocessing

The PhysioNet EEG Motor Movement/Imagery dataset was used for evaluation [14], [15]. Recordings were collected with the BCI2000 system [16]. The dataset contains 109 subjects with 64-channel EEG recordings during baseline, motor imagery, and motor execution tasks. The eyes-open baseline run R01 was used as the resting condition. Left/right motor imagery runs R04, R08, and R12 were used as the imagery condition, and left/right motor execution runs R03, R07, and R11 were used as the execution condition. Subjects 88, 92, and 100 were excluded because their baseline and task recordings had inconsistent sampling frequencies, leaving 106 subjects for paired analysis.

All recordings were loaded using MNE-Python [19]. EEG channel names were standardized, the standard 1005 montage was assigned, and all channels were band-pass filtered from 1 to 31 Hz. Baseline data were segmented into non-overlapping 4-s windows. Task data were epoched from 0 to 4 s following T1/T2 event annotations. All available task epochs from the selected runs were analyzed, and the baseline run yielded 15 non-overlapping 4-s windows per subject. Subject-level condition means were computed before statistical testing, so each subject contributed one paired value per condition. This produced paired subject-level measurements for rest, motor imagery, and motor execution under the same preprocessing and feature-extraction pipeline.

B. RSEC Shared-Energy Feature Extraction

The RSEC feature-extraction workflow is summarized in Fig. 1. Each EEG epoch was transformed into a complex analytic representation, mapped to a Riemann stereographic energy series, and summarized using shared-energy events across channel pairs.

Let $x_e(t)$ denote the EEG signal from channel e at time t within an epoch. The analytic signal is computed as

$$u_e(t) = x_e(t) + i\mathcal{H}\{x_e(t)\}, \quad (1)$$

where $\mathcal{H}\{\cdot\}$ is the Hilbert transform. For each channel and epoch, the analytic signal is normalized as

$$\eta_e(t) = \frac{u_e(t)}{a_e + \epsilon_n}, \quad (2)$$

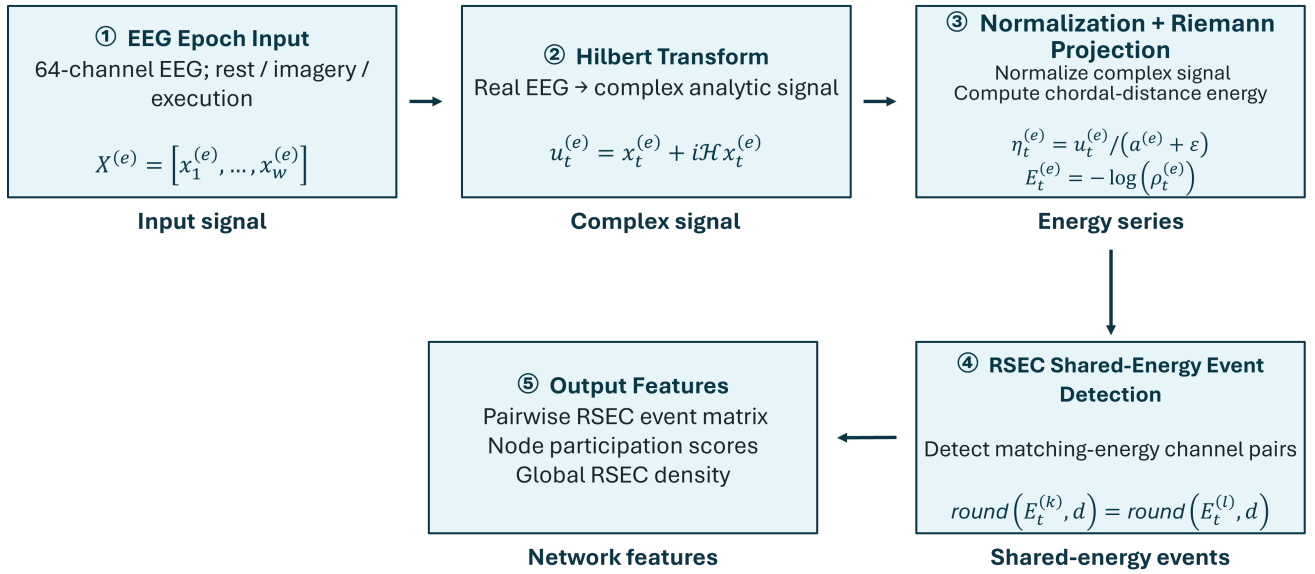


Fig. 1. Overview of the RSEC feature-extraction pipeline. Each multichannel EEG epoch is transformed into a complex analytic representation using the Hilbert transform. Channel-wise analytic signals are normalized within each epoch and mapped through the Riemann stereographic framework to obtain chordal-distance energy. Shared-energy events are detected between channel pairs and summarized as pairwise event matrices, node participation scores, and global RSEC density.

where $a_e = \max_t |u_e(t)|$ and ϵ_n is a small constant used to avoid division by zero. The normalized complex value is then mapped to a chordal-distance representation,

$$\rho_e(t) = \sqrt{\frac{1}{|\eta_e(t)|^2 + 1}}, \quad (3)$$

and the Riemann stereographic energy is computed as

$$E_e(t) = -\log(\rho_e(t)). \quad (4)$$

For each time point, a pair of channels (k, l) is counted as an RSEC shared-energy event when

$$\text{round}(E_k(t), d) = \text{round}(E_l(t), d), \quad (5)$$

where d is the fixed decimal precision used for event detection. In this implementation, $d = 3$. Global RSEC density is defined as

$$D = \frac{N_{\text{events}}}{T C(C-1)/2}, \quad (6)$$

where N_{events} is the total number of detected shared-energy events, T is the number of time samples in the epoch, and C is the number of EEG channels. Node participation scores were computed by counting how often each channel participated in shared-energy events and normalizing this count by the total number of possible events involving that channel.

Node participation is defined operationally as repeated channel involvement in RSEC shared-energy events. It should not be interpreted as direct evidence of anatomical hub structure without source-space localization and graph-theoretic validation.

C. Surrogate Control and Statistical Analysis

A phase-randomized surrogate control was used for the motor imagery condition. Surrogate data are commonly used to test whether a time-series effect remains after disruption of selected temporal structure [20]. For each EEG epoch and channel, phase randomization preserved the amplitude spectrum while disrupting temporal phase structure. RSEC features were recomputed on the surrogate signals using the same preprocessing, epoching, and event-detection parameters. This control tested whether the real motor-imagery task-minus-rest effect was reproduced after phase disruption.

For each subject, mean global RSEC density was computed separately for rest, motor imagery, and motor execution. Paired Wilcoxon signed-rank tests compared task and rest conditions. Task-minus-rest deltas were then compared between imagery and execution using a paired Wilcoxon signed-rank test. The real imagery task-minus-rest deltas were also compared with phase-randomized surrogate deltas. Cohen's d_z was reported as a within-subject standardized effect size. Statistical tests were planned a priori for four main contrasts: imagery versus rest, execution versus rest, imagery delta versus execution delta, and real imagery delta versus surrogate delta.

IV. RESULTS

Motor imagery produced a significant reduction in global RSEC density relative to rest in the 106-subject cohort (rest: 0.020489, imagery: 0.018230, mean delta: -0.002258 , Wilcoxon signed-rank $p = 0.00148$, Cohen's $d_z = -0.354$). In contrast, motor execution did not produce a significant global change relative to rest (rest: 0.020489, execution: 0.020487, mean delta: -0.000002 , $p = 0.367$, $d_z =$

TABLE I
PRIMARY GLOBAL RSEC DENSITY CONTRASTS. ALL TESTS ARE PAIRED
WILCOXON SIGNED-RANK TESTS ACROSS 106 SUBJECTS. Δ DENOTES
TASK MINUS REST.

Contrast	Δ	p	d_z
Imagery-rest	-0.002258	0.00148	-0.354
Execution-rest	-0.000002	0.367	-0.0003
Imagery-execution Δ	-0.002257	8.02×10^{-10}	-0.569

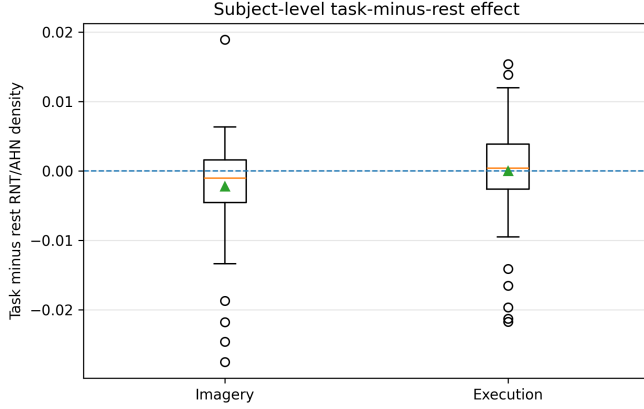


Fig. 2. Subject-level task-minus-rest global RSEC density changes for motor imagery and motor execution. Motor imagery showed a significant reduction relative to rest, while motor execution did not show a significant global-density change.

-0.0003). The task-minus-rest change differed significantly between motor imagery and motor execution (mean difference: -0.002257, $p = 8.02 \times 10^{-10}$, $d_z = -0.569$). The primary global-density contrasts are summarized in Table I, and the subject-level task-minus-rest distributions are shown in Fig. 2. These results indicate that the global RSEC response was motor-state specific rather than a generic rest-to-task effect.

The phase-randomized surrogate analysis tested whether the imagery-related reduction could be reproduced after disrupting temporal phase structure. In the real EEG data, motor imagery produced a significant task-minus-rest reduction in global RSEC density (mean delta: -0.002258, $p = 0.00148$). In contrast, the phase-randomized surrogate data produced a small, non-significant task-minus-rest change (mean delta: 0.000031, $p = 0.108$). The real and surrogate deltas differed significantly (mean difference: -0.002289, $p = 0.00153$), indicating that the real imagery-related decrease was not reproduced in direction or magnitude by the surrogate control. The surrogate statistics are reported in Table II, and the paired real-minus-surrogate differences are visualized in Fig. 3.

The largest node-participation changes during motor imagery were concentrated over frontopolar and frontal sensors, as shown in Fig. 4. These findings provide a sensor-space summary of where RSEC shared-energy participation changed most strongly. They should not be interpreted as anatomical source localization. Source-space analysis and region-level validation are reserved for future work.

TABLE II
PHASE-RANDOMIZED SURROGATE CONTROL FOR MOTOR IMAGERY. Δ
DENOTES TASK MINUS REST.

Contrast	Δ	p	d_z
Real imagery-rest	-0.002258	0.00148	-0.354
Surrogate imagery-rest	0.000031	0.108	0.137
Real-surrogate Δ	-0.002289	0.00153	-0.353

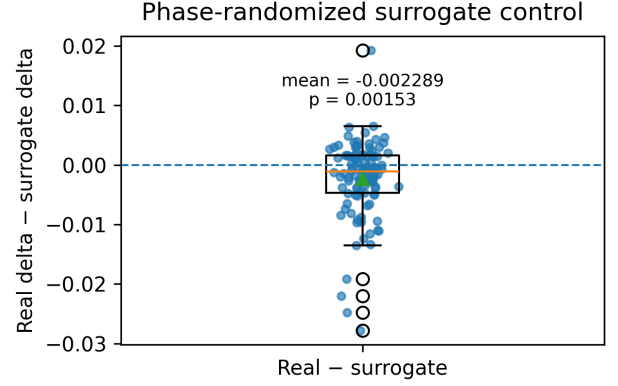


Fig. 3. Phase-randomized surrogate control for motor imagery. Each point represents one subject's real task-minus-rest delta minus the corresponding surrogate delta. Values below zero indicate a more negative real EEG effect. The real-minus-surrogate difference was significant ($p = 0.00153$), while the surrogate task-minus-rest effect alone was not significant ($p = 0.108$).

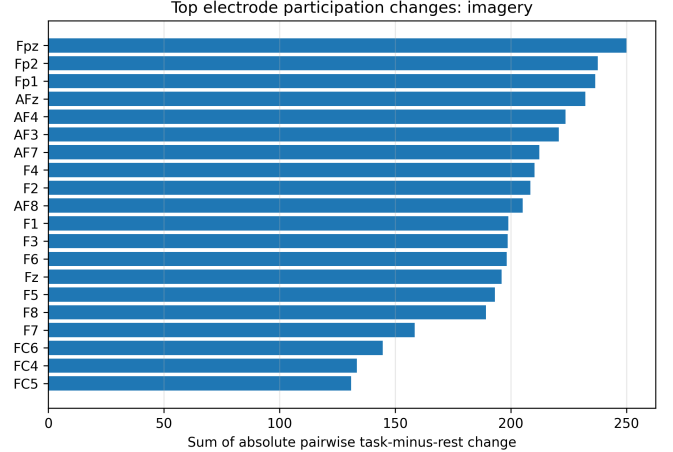


Fig. 4. Electrodes with the largest absolute task-minus-rest node-participation changes during motor imagery. Values summarize sensor-space participation changes and should not be interpreted as anatomical source localization.

V. DISCUSSION

The results support a conservative interpretation of RSEC as a sensor-space shared-energy feature. Global RSEC density appears to reflect state-dependent reorganization of EEG energy coupling rather than a simple increase or decrease associated with task engagement. As reported in Table I and Fig. 2, motor imagery reduced global RSEC density relative

to rest, whereas motor execution did not show a comparable global-density change. The significant imagery–execution contrast suggests that the proposed feature is sensitive to the type of motor state rather than only detecting a generic change from rest to task.

One possible interpretation is that motor imagery shifts the EEG signal away from broad resting-state shared-energy coupling toward a more selective internally generated motor state. Motor execution may produce a different global outcome because overt movement combines motor commands, proprioceptive feedback, and sensory consequences. These additional processes may change regional or pairwise patterns without producing the same net reduction in global RSEC density.

The phase-randomized surrogate analysis provides an additional control. As shown in Table II and Fig. 3, the surrogate data produced a small, non-significant positive task-minus-rest change and did not reproduce the direction or magnitude of the real imagery-related reduction. This finding indicates that the real motor-imagery effect depends on temporal structure disrupted by phase randomization.

The node-participation results should also be interpreted cautiously. The largest changes were observed over frontopolar and frontal sensors, but these results are sensor-space summaries only. They should not be interpreted as anatomical source localization or as direct evidence of anatomical hub structure. In this paper, node participation means repeated involvement in RSEC-defined shared-energy events at the sensor level.

This study focuses on the left/right motor imagery and execution paradigm, broadband 1–31 Hz preprocessing, and the original fixed-precision RSEC shared-energy definition. These choices keep the analysis focused and reproducible, but they also define the scope of the present study. Future studies should examine frequency-specific RSEC behavior in sensorimotor bands, alternative event definitions that reduce dependence on fixed decimal precision, and spatially informed summaries that reduce the influence of short-distance sensor pairs. Additional validation should include stronger surrogate models, comparison with standard EEG spectral and connectivity features, source-space analysis, and classification experiments.

VI. CONCLUSION

This paper presented Riemann Stereographic Energy Coupling (RSEC) as a nonlinear sensor-space EEG feature for motor-state analysis. In 106 usable subjects from the PhysioNet EEG Motor Movement/Imagery dataset, motor imagery was associated with a significant reduction in global RSEC density relative to rest. A comparable global-density change was not observed during motor execution. The imagery-related contrast also differed from both the execution contrast and phase-randomized surrogate EEG. Overall, the results suggest that RSEC is sensitive to motor-imagery-related changes in sensor-space EEG organization. Future work should examine band-specific versions of the method, additional control analy-

ses, source-space extensions, and comparisons with established EEG connectivity measures.

DATA AND CODE AVAILABILITY

A reproducibility package accompanies this manuscript. The code repository includes scripts for downloading the PhysioNet EEG Motor Movement/Imagery dataset through MNE-Python, computing RSEC shared-energy features, reproducing the reported statistics, and generating the figures: <https://github.com/bayazitkaraman/Riemann-Stereographic-Energy-Coupling>. Raw EEG recordings are not redistributed in the repository.

REFERENCES

- [1] G. Pfurtscheller and F. H. Lopes da Silva, “Event-related eeg/meg synchronization and desynchronization: Basic principles,” *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842–1857, 1999.
- [2] G. Pfurtscheller and C. Neuper, “Motor imagery and direct brain-computer communication,” *Proceedings of the IEEE*, vol. 89, no. 7, pp. 1123–1134, 2001.
- [3] P. L. Nunez, R. Srinivasan, A. F. Westdorp, R. S. Wijesinghe, D. M. Tucker, R. B. Silberstein, and P. J. Cadusch, “Eeg coherency: I. statistics, reference electrode, volume conduction, laplacians, cortical imaging, and interpretation at multiple scales,” *Electroencephalography and Clinical Neurophysiology*, vol. 103, no. 5, pp. 499–515, 1997.
- [4] G. Nolte, O. Bai, L. Wheaton, Z. Mari, S. Vorbach, and M. Hallett, “Identifying true brain interaction from eeg data using the imaginary part of coherency,” *Clinical Neurophysiology*, vol. 115, no. 10, pp. 2292–2307, 2004.
- [5] C. J. Stam, G. Nolte, and A. Daffertshofer, “Phase lag index: Assessment of functional connectivity from multi channel eeg and meg with diminished bias from common sources,” *Human Brain Mapping*, vol. 28, no. 11, pp. 1178–1193, 2007.
- [6] C. Neuper, R. Scherer, M. Reiner, and G. Pfurtscheller, “Imagery of motor actions: Differential effects of kinesthetic and visual-motor mode of imagery in single-trial eeg,” *Cognitive Brain Research*, vol. 25, no. 3, pp. 668–677, 2005.
- [7] E. Bullmore and O. Sporns, “Complex brain networks: Graph theoretical analysis of structural and functional systems,” *Nature Reviews Neuroscience*, vol. 10, no. 3, pp. 186–198, 2009.
- [8] M. Rubinov and O. Sporns, “Complex network measures of brain connectivity: Uses and interpretations,” *NeuroImage*, vol. 52, no. 3, pp. 1059–1069, 2010.
- [9] M. P. van den Heuvel and O. Sporns, “Network hubs in the human brain,” *Trends in Cognitive Sciences*, vol. 17, no. 12, pp. 683–696, 2013.
- [10] J.-P. Lachaux, E. Rodriguez, J. Martinerie, and F. J. Varela, “Measuring phase synchrony in brain signals,” *Human Brain Mapping*, vol. 8, no. 4, pp. 194–208, 1999.
- [11] M. Vinck, R. Oostenveld, M. van Wingerden, F. Battaglia, and C. M. A. Pennartz, “An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias,” *NeuroImage*, vol. 55, no. 4, pp. 1548–1565, 2011.
- [12] B. Karaman, R. M. Demirer, C. Bayrak, and M. M. Su, “Modeling the antipodal connectivity structure of neural communities,” *AIMS Neuroscience*, vol. 3, no. 2, pp. 163–180, 2016.
- [13] B. Karaman, “The role of antipodal connectivity structure on cognitive functions and mental disorders,” Ph.D. dissertation, University of Arkansas at Little Rock, Little Rock, AR, USA, 2018.
- [14] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, “Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [15] G. Schalk, “EEG Motor Movement/Imagery Dataset,” <https://physionet.org/content/eegmmidb/1.0.0/>, 2009, physioNet, version 1.0.0.
- [16] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, “Bci2000: A general-purpose brain-computer interface (bci) system,” *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004.

- [17] T. Needham, *Visual Complex Analysis*. Oxford, UK: Oxford University Press, 1997.
- [18] E. B. Saff and A. D. Snider, *Fundamentals of Complex Analysis for Mathematics, Science, and Engineering*. Upper Saddle River, NJ, USA: Prentice Hall, 1997.
- [19] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, and M. Hämäläinen, “MEG and EEG data analysis with MNE-Python,” *Frontiers in Neuroscience*, vol. 7, p. 267, 2013.
- [20] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, and J. D. Farmer, “Testing for nonlinearity in time series: The method of surrogate data,” *Physica D: Nonlinear Phenomena*, vol. 58, no. 1–4, pp. 77–94, 1992.